CHANDIGARH UNIVERSITY

UNIVERSITY INSTITUTE OF COMPUTING (UIC)

BACHELOR OF COMPUTER APPLICATIONS (BCA)

PROJECT REPORT ON

CLOUD-BASED STUDENT LEARNING MANAGEMENT SYSTEM (LMS)

Submitted in partial fulfilment of the requirements for the degree of Bachelor of Computer Applications (BCA)

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CERTIFICATE

This is to certify that the project report entitled **"Cloud-Based Student Learning Management System (LMS)"** has been prepared and submitted by Rohit (UID: 24BCA10303) and Rohit (UID: 24BCA10410), students of the Bachelor of Computer Applications (BCA) programme at the University Institute of Computing, Chandigarh University.

The project report has been prepared under the guidance of the assigned project supervisor and is submitted in partial fulfilment of the academic requirements of the BCA programme.

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DECLARATION

We hereby declare that the project report entitled **"Cloud-Based Student Learning Management System (LMS)"** is prepared by us as part of our academic project work for the Bachelor of Computer Applications programme at Chandigarh University.

We declare that the information presented in this report is prepared for academic purposes and that all external references used have been acknowledged appropriately.

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We also acknowledge the contribution of our classmates, friends and family members for their encouragement and support.

This project has helped us understand the practical applications of web technologies, database management and cloud computing in the education sector.

ABSTRACT

The Cloud-Based Student Learning Management System (LMS) is a proposed web-based application designed to improve the process of online learning and academic management.

The system aims to provide a common platform where students can access study materials, view announcements, submit assignments and monitor their academic activities. Teachers can upload learning resources, share important information, create assignments and review student submissions. Administrators can manage users, courses and system information.

The proposed system uses cloud-based infrastructure to make educational resources accessible through internet-connected devices. It aims to reduce manual work, improve communication between students and teachers, and provide an organised learning environment.

The project is planned using technologies such as Python, Flask, HTML, CSS, JavaScript and a relational database such as SQLite or MySQL. The final technology selection and deployment will depend on implementation requirements.

Keywords: Cloud Computing, Learning Management System, Online Education, Student Portal, Database Management, Web Application.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

Education is increasingly supported by digital technologies. Students and teachers use online platforms to communicate, share educational resources and manage academic activities.

A Learning Management System (LMS) is a software application that helps educational institutions organise and manage learning activities through a digital platform.

The proposed Cloud-Based Student Learning Management System provides a centralised platform for students, teachers and administrators. It is intended to make learning resources available online and simplify academic communication.

1.2 Background

Traditional learning management often involves physical documents, manual assignment collection and communication through different channels. These methods can make academic information difficult to organise.

A cloud-based LMS can provide a common online environment where educational resources and academic information can be managed more efficiently.

1.3 Problem Statement

Students may face difficulties accessing study materials, receiving timely announcements and submitting assignments when academic information is distributed across different platforms.

Teachers may also spend additional time managing assignments, sharing resources and maintaining student records manually. Therefore, a centralised online learning platform is proposed to simplify these activities.

CHAPTER 2: OBJECTIVES AND SCOPE

2.1 Objectives

The main objectives of the proposed project are:

1. To develop a centralised online learning platform.
2. To provide secure login facilities for students, teachers and administrators.
3. To allow teachers to upload and manage study materials.
4. To enable students to access learning resources anytime through the internet.
5. To provide online assignment submission facilities.
6. To improve communication through announcements and notifications.
7. To maintain organised information about users and courses.

2.2 Scope

The proposed system will include:

- Student registration and login.
- Teacher and administrator login.
- Course and subject management.
- Uploading and downloading study materials.
- Assignment creation and submission.
- Announcements and notifications.
- Basic student progress and academic activity tracking.

The initial version will focus on essential academic functions. Advanced features may be introduced in future versions.

2.3 Feasibility Study

Technical Feasibility: The project can be developed using commonly available web development technologies such as Python, Flask, HTML, CSS and JavaScript.

Economic Feasibility: Open-source development tools and database systems can reduce initial development costs.

Operational Feasibility: The interface is intended to be simple so that students, teachers and administrators can use the system with basic computer knowledge.

Schedule Feasibility: The project can be developed in stages, beginning with user authentication and followed by course, material and assignment modules.

CHAPTER 3: REQUIREMENTS AND TECHNOLOGY STACK

3.1 Hardware Requirements

- Computer or laptop.
- Minimum 4 GB RAM recommended.
- Internet connection.
- Sufficient storage for application files and learning materials.

3.2 Software Requirements

- Operating System: Windows/Linux.
- Code Editor: Visual Studio Code.
- Web Browser: Google Chrome or Microsoft Edge.
- Python 3.
- Flask framework.
- Database: SQLite or MySQL.

3.3 Technology Stack

Frontend: HTML, CSS and JavaScript.

Backend: Python with Flask.

Database: SQLite or MySQL.

Cloud Hosting: A suitable cloud hosting service may be selected during deployment.

Development Tool: Visual Studio Code.

3.4 Functional Requirements

1. The system should allow authorised users to log in.
2. Students should be able to view available courses.
3. Teachers should be able to upload study materials.
4. Students should be able to download available resources.
5. Teachers should be able to create assignments.
6. Students should be able to submit assignments online.
7. Administrators should be able to manage users and courses.

3.5 Non-Functional Requirements

- **Security:** User information should be protected.
- **Usability:** The interface should be easy to understand.
- **Reliability:** The application should handle user requests properly.
- **Performance:** Pages should load efficiently.
- **Scalability:** The design should allow future expansion.
- **Availability:** The cloud-based system should aim for access through the internet.

CHAPTER 4: SYSTEM ARCHITECTURE AND DESIGN

4.1 System Architecture

The proposed LMS follows a three-layer architecture:

1. **Presentation Layer:** Contains the user interface developed using HTML, CSS and JavaScript.
2. **Application Layer:** Processes user requests, handles authentication and performs application operations using Python and Flask.
3. **Database Layer:** Stores user details, course information, assignments, submissions and learning resources.

Architecture Diagram

```
STUDENTS / TEACHERS / ADMIN | v WEB BROWSER | v PRESENTATION LAYER (HTML, CSS, JavaScript) |
v APPLICATION LAYER (Python Flask) | v DATABASE LAYER (SQLite / MySQL) | v CLOUD HOSTING
SERVER
```

4.2 Data Flow Diagram (DFD)

Level 0: Context Diagram

```
STUDENT ----------------------+ | v +-------------+ | | TEACHER -------------->| LMS | | |
ADMIN ---------------->| | +-------------+ | v DATABASE AND STORAGE
```

Level 1: Main Processes

```
User Login ---> Authentication ---> Dashboard | +------> Course Management +------> Study
Material Management +------> Assignment Management +------> Notifications | v Database
```

4.3 User Roles

Student: Can access courses, view study materials, submit assignments and check announcements.

Teacher: Can manage courses, upload study materials, create assignments and review submissions.

Administrator: Can manage user accounts, courses and overall system information.

CHAPTER 5: MODULES OF THE SYSTEM

5.1 User Authentication Module

This module manages user login and access control. Users will log in using their registered credentials. Access to features will depend on the assigned role.

5.2 Student Module

The student module provides access to: Student dashboard, Enrolled courses, Study materials, Assignment details, Assignment submission, and Announcements.

5.3 Teacher Module

Teachers will be able to manage assigned courses, upload study materials, create assignments, view student submissions, and provide feedback and marks.

5.4 Administrator Module

The administrator will manage student accounts, teacher accounts, course information, user access, and general system information.

5.5 Study Material Module

This module allows teachers to upload documents and other learning resources. Students can view and download the available materials.

5.6 Assignment Module

Teachers can create assignments with descriptions and submission deadlines. Students can submit their work through the platform.

CHAPTER 6: DATABASE DESIGN

The database is an important component of the proposed LMS. It stores and manages information related to users, courses, study materials and assignments.

6.1 Proposed Database Tables

1. Users Table

Field	Description
user_id	Unique user ID
name	User name
email	User email

Field	Description
password_hash	Encrypted password hash
role	Student, Teacher or Admin

2. Courses Table

Field	Description
course_id	Unique course ID
course_name	Name of course
description	Course details
teacher_id	Assigned teacher

3. Materials Table

Field	Description
material_id	Unique material ID
course_id	Related course
title	Material title
file_path	Stored file location
upload_date	Upload date

4. Assignments Table

Field	Description
assignment_id	Unique assignment ID
course_id	Related course
title	Assignment title
description	Assignment details
deadline	Submission deadline

5. Submissions Table

Field	Description
submission_id	Unique submission ID
assignment_id	Related assignment
student_id	Student ID
file_path	Submitted file location
submission_date	Submission date

6.2 Database Relationships

- One teacher can manage multiple courses.
- One course can contain multiple study materials.
- One course can contain multiple assignments.
- One assignment can receive multiple student submissions.
- Each submission belongs to a particular student and assignment.

CHAPTER 7: IMPLEMENTATION AND TESTING

7.1 Implementation Plan

1. Requirement analysis and project planning.
2. Designing the user interface.
3. Developing the backend using Python Flask.
4. Creating database tables.
5. Developing authentication and role-based access.
6. Implementing course and study material management.
7. Developing assignment and submission facilities.
8. Testing the application.
9. Preparing for cloud deployment.

7.2 Testing Strategy

Testing will be performed to verify that the application works according to its requirements.

Test Case	Expected Result
Valid login	User dashboard opens
Invalid login	Error message appears
Course access	Course details are displayed
Material upload	Resource is stored

Test Case	Expected Result
Material download	Student can access resource
Assignment creation	Assignment is displayed
Assignment submission	Submission is recorded
Role verification	Correct permissions are applied

Note: These are proposed test cases. Actual test results should be entered after the application has been developed and tested.

CHAPTER 8: ADVANTAGES, LIMITATIONS AND FUTURE SCOPE

8.1 Advantages

1. Centralised management of educational resources.
2. Easy access to learning materials.
3. Improved communication between students and teachers.
4. Reduced paperwork.
5. Online assignment management.
6. Better organisation of academic information.
7. Potential access from different internet-connected devices.

8.2 Limitations

1. An internet connection is required for online access.
2. Cloud hosting may involve operational costs.
3. Large files may require additional storage.
4. User data must be protected through proper security measures.
5. The initial version may not include advanced learning analytics.

8.3 Future Scope

- Live online classes.
- Video lectures.
- Online quizzes and examinations.
- Automated attendance management.
- AI-based personalised learning recommendations.
- Mobile application support.
- Performance analytics and progress reports.
- Integration with payment and university management systems where required.

CONCLUSION

The Cloud-Based Student Learning Management System (LMS) is a proposed solution for organising educational activities through a centralised online platform.

The system aims to provide students with easy access to learning resources and assignment facilities while helping teachers manage courses and academic content. The administrator module will support user and course management.

By using web development technologies, database management and cloud infrastructure, the proposed system can provide a flexible foundation for digital learning.

This project also provides practical learning opportunities in Python programming, web development, database design, software engineering and cloud computing.

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